Derivative-point case: an actual weighted-Proj DVR

Formal proof packet

28 October 2026

This packet records a checked local geometry theorem. It contains no experiment, target result, or cost ratio. The Lean source, compiler log, editable SVG, and CSV gate table accompany this PDF.

Checked boundary

Let K be any field, $f(X) = X^5 + a_3X^3 + a_2X^2 + a_1X + a_0$, and $(a, y) \in K^2$ satisfy $y^2 = f(a)$. Assume f is separable and $f'(a) \neq 0$. The difference quotient $q_a(X) = (f(X) - f(a))/(X - a)$ has $q_a(a) = f'(a)$, so it is a unit in the local ring at (a, y). The checked point ideal is $(X - a, Y - y)$, and the curve relation gives

$$(Y - y)(Y + y) = (X - a)q_a(X).$$

Hence the local maximal ideal is generated by $Y - y$. Its generator is nonzero. Separability supplies the coordinate-ring domain, and the checked DVR criterion makes the affine localization a discrete valuation ring. The checked ring equivalence transfers this structure to the *actual weighted*-Proj *point stalk*. The theorem works in characteristic two when this pointwise derivative condition holds; it does not assume $2y \neq 0$.

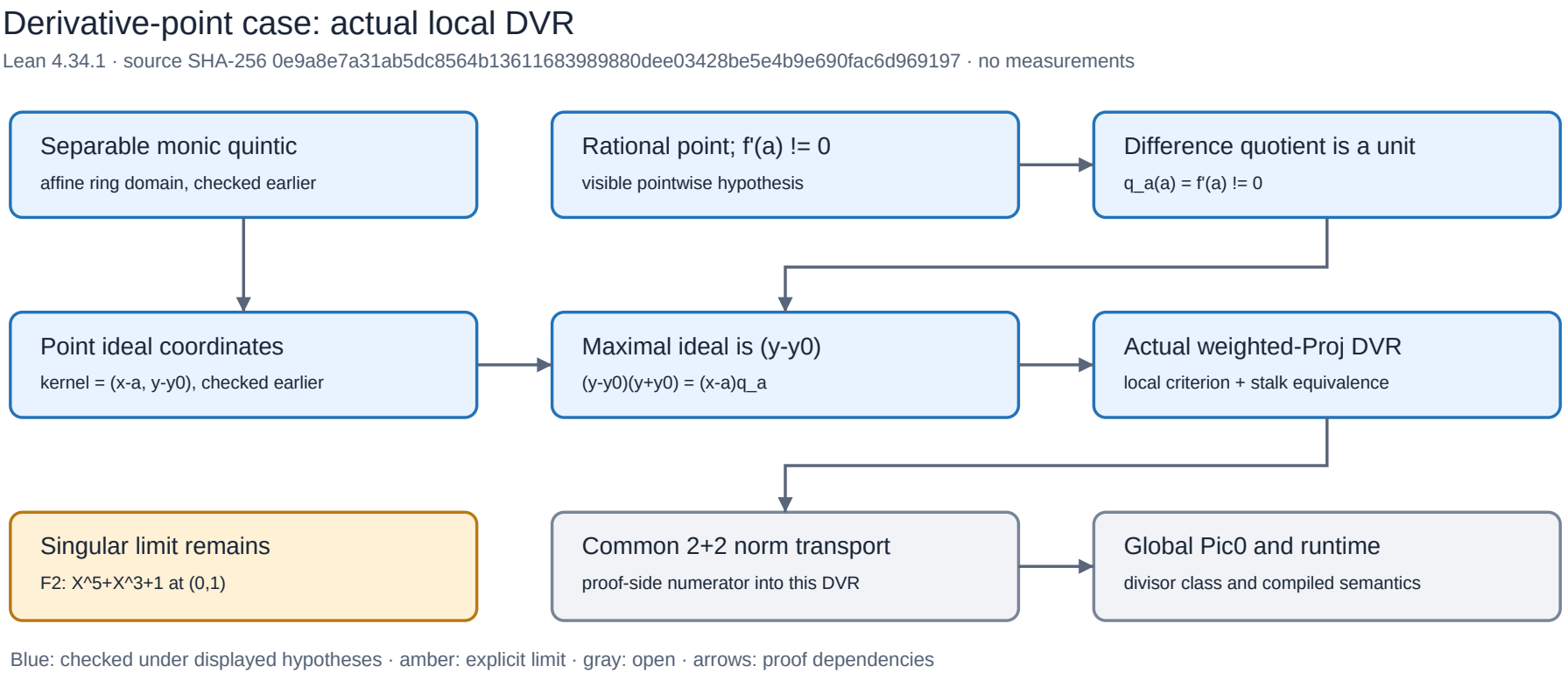
Scope and verification

No DVR claim is made when both partial derivatives vanish. Over $\mathbf{F}_2$, $f = X^5 + X^3 + 1$ is separable but $(0, 1)$ is singular. A combined theorem for the two pointwise smoothness cases remains open in this formal line.

Direct pinned Lean 4.34.1 compilation exited zero against Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`. Five printed axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Source SHA-256: `0e9a8e7a31ab5dc8564b13611683989880dee03428be5e4b9e690fac6d969197`. The compiler log retains nonfatal style-linter suggestions. The local DVR criterion follows pinned Mathlib `IsDiscreteValuationRing.TFAE` and Stacks Project Tag 00PD. The prior checked stalk equivalence uses the open-immersion principle of Stacks Tag 01HD.

Formula-specific common 2+2 norm transport, global principal-divisor/Pic^0 transport, and compiled Rust semantics remain open. The CSV records exact gate statuses.

Proof dependencies and open gates



Blue nodes are checked under the displayed hypotheses. Amber records the singular limit; gray nodes are open. Arrows show proof dependencies, not measured performance.